Doppler Ultrasound

Doppler Effect

The Doppler effect is the change in frequency (or wavelength) of a sound wave when there is relative motion between the source and the observer.

- When the source approaches → frequency increases (sound becomes higher)
- When the source moves away → frequency decreases (sound becomes lower)

When does it occur?

It occurs when there is relative motion between:

- Sound source
- Observer
- Medium (e.g., air)

Basic idea

- Approaching → waves become closer → higher frequency
- Moving away → waves become farther → lower frequency

Main cases of Doppler Effect

1. Moving source, stationary observer

2. Moving observer, stationary source

3. Both source and observer are moving

General formula (simplified)

$$\acute{f} = f \times \frac{(c \pm v_{observer})}{(c \mp v_{source})}$$

Where:

- ($\acute{f}$) = observed frequency
- (f) = real frequency
- (**c**) = speed of sound
- ($v_{observer}$) = observer speed
- (v_{source}) = source speed

Examples

- Ambulance siren changing sound as it passes
- Train whistle

Medical applications

- Measuring blood flow
- Checking heart valves
- Detecting blocked blood vessels

Problem 1:

An ambulance produces a sound with a real frequency of (f = 500 , Hz).
The ambulance is moving **towards a stationary observer** at a speed of (v_{source} = 20 , m/s).
The speed of sound in air is (c = 340 , m/s).

Calculate the frequency heard by the observer.

Solution:

$$\acute{f} = f \times \frac{(c \pm v_{observer})}{(c \mp v_{source})}$$

Given:

- Observer is stationary → ($v_{observer}$ = 0)
- Source is approaching → use minus sign in the denominator

$'\acute{f} = 500 * \frac{340}{340-20}$

$\acute{f} = 500 * \frac{340}{320}$

$\acute{f} = 500 * 1.0625$

$\acute{f} = 531.25$, Hz

Problem2 :

A train whistle produces a sound with a real frequency of (f = 600 , Hz).
A person is moving **towards the stationary source** at a speed of ($v_{observer}$ } = 30 , m/s).
The speed of sound in air is (c = 340 , m/s).

Calculate the frequency heard by the observer.

Solution:

$$\acute{f} = f \times \frac{(c \pm v_{observer})}{(c \mp v_{source})}$$

Given:

- Source is stationary → ($v_{source} = 0$)
- Observer is moving **towards the source** → use plus sign in numerator

$\acute{f} = 600 * \frac{340+30}{340}$

$\acute{f} = 600 * \frac{370}{340}$

$\acute{f} = 500 * 1.0882$

$\acute{f} = 652.9$, Hz

Continuous and Pulsed Doppler

Continuous Doppler

Continuous Doppler is a type of ultrasound Doppler in which sound waves are continuously transmitted and received without interruption.

Characteristics:

- Measures high blood flow velocities accurately
- Cannot determine the exact depth (location) of the signal
- Used when blood flow is very fast

Uses:

- Measuring blood flow in large vessels
- Evaluating heart valves

Pulsed Doppler

Pulsed Doppler is a type of ultrasound Doppler that sends sound waves in **short pulses and receives echoes between them**.

Characteristics:

- Can measure blood flow at a specific depth (location)
- Useful for targeting a specific point in a vessel
- Less accurate for very high blood flow velocities

Uses:

- Examining specific blood vessels
- Measuring blood flow at a chosen location in the body

Blood Flow Measurements

Blood flow measurement is the process of measuring the speed and volume of blood moving through blood vessels in the human body. It helps doctors understand how well blood is circulating.

Why is it important?

It is used to:

- Check heart function
- Detect blocked or narrowed blood vessels
- Evaluate organ health (brain, kidneys, lungs)
- Monitor patients with cardiovascular diseases

Methods used to measure blood flow

1. Doppler Ultrasound

The most common method.

- Uses sound waves to measure blood movement
- Based on the Doppler effect
- Non-invasive and safe

It can measure:

- Blood velocity
- Direction of flow

2. Angiography

- Uses X-rays and contrast dye

- Shows detailed images of blood vessels
- Helps detect blockages or narrowing

3. MRI Angiography (MRA)

- Uses magnetic resonance imaging
- No radiation
- Provides detailed images of blood vessels

4. CT Angiography

- Uses CT scans with contrast material
- Gives fast and clear images of blood flow

What parameters are measured?

- Blood flow velocity (speed)
- Blood flow volume (amount per time)
- Direction of flow
- Pressure changes in vessels

Clinical Doppler Applications

Doppler ultrasound is widely used in medicine to measure and analyze blood flow in the body. It helps doctors diagnose many cardiovascular and vascular conditions in a safe, non-invasive way.

1. Cardiology

Used to evaluate heart function and blood flow through the heart.

Applications:

- Assess heart valve diseases (stenosis or regurgitation)
- Measure blood flow across heart chambers
- Detect congenital heart defects
- Evaluate heart pumping efficiency

2. Vascular System

Used to examine arteries and veins in the body.

Applications:

- Detect arterial stenosis (narrowing)
- Identify blood clots (deep vein thrombosis – DVT)
- Evaluate peripheral artery disease (PAD)
- Assess carotid artery disease (risk of stroke)

3. Obstetrics

Used to monitor blood flow in pregnancy.

Applications:

- Check blood flow in the umbilical cord
- Assess fetal well-being
- Detect placental insufficiency
- Monitor fetal growth and oxygen supply

4. Neurology

Used to study blood flow in brain vessels.

Applications:

- Detect stroke risk
- Evaluate cerebral blood circulation
- Identify vascular abnormalities in the brain

5. Abdominal Organs

Used to assess blood flow in internal organs.

Applications:

- Liver blood flow evaluation (portal hypertension)
- Kidney perfusion assessment
- Detect tumors with abnormal blood supply

6. Emergency Medicine

Used for quick assessment in critical cases.

Applications:

- Detect internal bleeding

- Evaluate trauma patients

Medical Applications of Ultrasound: B-mode, M-mode, and Real-Time Imaging

Ultrasound imaging is a widely used diagnostic tool that uses high-frequency sound waves to create images of internal body structures. Different modes are used depending on the clinical purpose.

1. B-mode (Brightness Mode)

B-mode produces a **2D grayscale image** where:

- Brightness represents the strength of the echo
- Structures appear as two-dimensional anatomical images

Medical Applications

- Abdominal organ imaging (liver, kidneys, spleen)
- Pregnancy and fetal imaging
- Thyroid and breast examination
- Detecting tumors and cysts

Why it is important?

- Provides clear anatomical details
- Most commonly used ultrasound mode in hospitals

2. M-mode (Motion Mode)

M-mode shows **movement over time** in a single line:

- Displays motion of structures as a graph
- Very high temporal resolution

Medical Applications

- Cardiac imaging (echocardiography)
- Measuring heart wall motion
- Evaluating heart valve movement
- Fetal heart rate monitoring

- Assessing heart function (left ventricle motion)

Why it is important?

- Very accurate for studying fast motion like heartbeats

3. Real-Time Imaging (2D/3D Live Ultrasound)

Real-time ultrasound provides continuous live images of moving organs.

Medical Applications

- Guiding surgical procedures (biopsies, needle insertion)
- Monitoring fetal movements during pregnancy
- Cardiac function assessment in real time

Why it is important?

- Allows doctors to see movement instantly
- Helps in safe and accurate procedures

Mode	Main Feature	Key Medical Uses
B-mode	2D anatomical images	Organs, pregnancy, tumors
M-mode	Motion over time (graph)	Heart motion, fetal heart rate
Real-time	Live moving images	Procedures, pregnancy, emergency care

Ultrasound Imaging Systems

What is an Ultrasound Imaging System?

An ultrasound imaging system is a medical device that uses high-frequency sound waves to create images of internal body structures. It is widely used because it is safe, real-time, and non-invasive.

Main Components of an Ultrasound System

- **Transducer (Probe):** Sends and receives ultrasound waves

- **Main Processing Unit:** Processes echoes into images
- **Display Monitor:** Shows the ultrasound image
- **Control Panel:** Adjusts settings (frequency, depth, gain)
- **Power Supply:** Provides energy to the system

Image Artifacts

What are artifacts?

Artifacts are false or misleading structures that appear in ultrasound images but do not actually exist in the body.

Common Types of Artifacts

1. Acoustic Shadowing

2. Enhancement (Increased Echo)

Safety in Ultrasound

Is ultrasound safe?

Yes, ultrasound is generally considered **very safe** because it does not use ionizing radiation like X-rays or CT scans.

Safety Principles

- Use the **lowest possible energy** (ALARA principle: As Low As Reasonably Achievable)
- Limit exposure time
- Use correct frequency for the examination
- Proper training of operators

Industrial and Scientific Applications of Ultrasound

Ultrasound is not only used in medicine; it also has many important **industrial and scientific applications** because of its ability to travel through materials and reflect from boundaries.

Summary Table

Application	Main Idea	Uses

Application	Main Idea	Uses
NDT	Detect internal defects without damage	Aircraft, pipelines
SONAR	Uses sound waves in water	Navigation, submarines
Ultrasonic Cleaning	Removes dirt using cavitation	Instruments, jewelry
Ultrasonic Welding	Joins materials using vibration	Plastics, electronics

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